

Risk and safety in the access network

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This paper discusses an initiative to reduce risk in the access network by better management of the pole stock. The initiative concerns the pole renewal programme and proposes a risk assessment-based prioritisation methodology which enables remedial activity to be targeted on the basis of highest risk.

1. Introduction

As part of its access distribution technology, BT operates an extensive overhead network, comprising some 4 million poles and 20 million wires. Ensuring this network is safe, that it poses minimum risk to BT people and the general public, and complies with the company's legal and regulatory obligations, is the responsibility of the asset assurance programme. This programme constitutes considerable expenditure with ongoing annual funding of the order of £100 million, of which some two thirds is spent on the overhead network. It is therefore essential to ensure that the company is getting good value for money and that the spend is being effectively targeted where it will bring the most benefit. To this end considerable effort has been invested in evolving and refining the programme strategy for work prioritisation and ensuring the understanding and management of the risks posed by operating such a network.

The response to real and perceived safety hazards in the past has been volume-based rather than risk-based. Although, as a company, BT reacts very quickly to dangerous situations, the general practice has been to work through the backlog of safety defects on a geographic basis. It is now proposed to move to a risk-based methodology and the D¹ pole renewal sub-programme, at approximately £40 million per year, provides a useful vehicle to explain this approach.

2. A case study in risk assessment — D poles

Information on this type of defect is obtained from two sources — proactively from the periodic pole test cycle (carried out by dedicated pole testers), or *ad hoc* from engineers working in the overhead network and identifying defects. The high level process is shown in Fig 1.

Pole test data is currently captured using a variety of methods. The majority is identified by the dedicated pole

testers and they, for the most part, use a hand-held terminal (HHT) with data packs being posted to their control on a weekly basis.

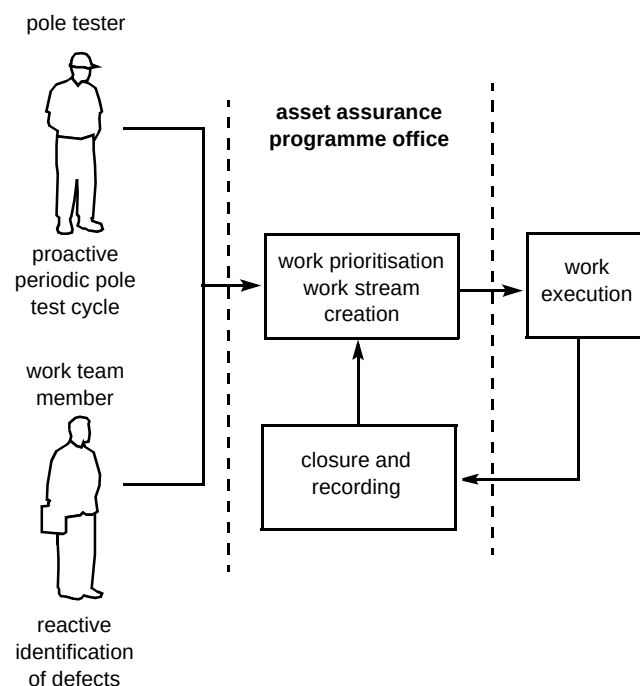


Fig 1 The process of pole renewal.

Over the last few years there has been a significant increase in the number of poles being found defective during the periodic pole test cycle. The national 'find rate' has increased dramatically with one geographic area showing a near fourfold increase. The majority of this increase has been due to the identification of poles that are not set deep enough in the ground, i.e. shallow. Most of these poles are between 0 to 0.5 metres (nominally 20 inches) below the standard; the distribution is shown in Fig 2. These poles may have previously passed a number of periodic pole tests and been climbed repeatedly. This historical evidence suggests that the

¹ A D pole is a pole which has been identified as defective by a pole tester. A pole may be marked D for a variety of reasons — damage, decay, not being set at the correct depth, etc.

depth criterion is not the absolute determinant when considering the risk of failure of a pole. Despite this, these poles have been scheduled for replacement and attract the same priority as poles which are in more urgent need of attention.

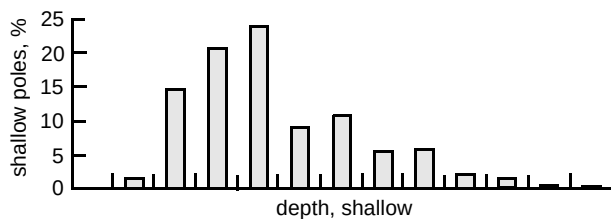


Fig 2 The distribution of shallowness for poles below the nominal depth.

A selection of experienced pole testers was interviewed to try to identify the cause of this sudden increase in the find rate. The increase coincided with the introduction of field manager quality checks and the introduction of the original hand-held terminal (referred to above). Prior to these changes, pole testers had been carrying out informal risk assessments using their expertise and experience — assessing poles as safe even when they did not exactly meet the depth requirement. This unrecorded practice ceased following the introduction of the quality checks and the hand-held terminal, thus triggering the increase in the find rate. Set against this increased find rate is the fact that the actual injury rate from dangerous poles has remained small, with no recorded incidence of death and few of serious injury to members of the public or BT people. The pole population had not therefore become noticeably less safe. The historical evidence suggests that these experience-based risk assessments were both accurate and valid.

Associated with the above is the problem of a lack of effective management information to enable programme performance to be maximised. Neither of the pole test data recording methodologies enable anything other than a very basic prioritisation technique to be applied.

3. The risk assessment approach

The management of Health and Safety at Work Regulations (1992) requires employers to carry out suitable and sufficient risk assessment of work activities. This includes identifying potential hazards, determining the severity of possible injury to people and then taking appropriate steps to eliminate that risk as far as is reasonably practicable.

Safety consultancy services have carried out risk assessments on behalf of BT and it was sensible to use existing expertise within the company to assist in a project commissioned to resolve the problem areas identified. The following key steps were taken:

- identify the informal risk assessment criteria and the process used by experienced pole testers,
- from this, develop a formal risk assessment methodology,
- capture that methodology in an algorithm,
- identify and procure a new more powerful hand-held terminal,
- design a new bespoke database with a powerful reports package,
- programme the algorithm into a new hand-held terminal.

From this a number of benefits can be realised:

- the D pole find rate brought back towards the figures of recent years,
- risk assessment is formalised and controlled, with the company accepting responsibility rather than individual pole testers,
- the development of a more powerful management information system to support field managers in pole tester coaching,
- prioritisation of those poles that require more urgent attention as they should no longer be masked by marginally shallow poles.

There are several methodologies for calculating risk, for example:

- site visits and investigation,
- anecdotal evidence based on individuals' experiences,
- empirical examination of the pole testers' reports.

The methodology chosen was empirical examination of reports. This choice was made for the following reasons:

- information already existed,
- the methodology complies with the law,
- the methodology is robust and can withstand critical scrutiny,
- it is cost effective.

The output of this examination was a list of factors and related weightings which determine the risk of failure of a pole, e.g. loading, lean, depth. These factors are then plugged into a risk estimation model to arrive at a hazard rating number (HRN). The model uses the following formula:

$$\text{HRN} = \text{FE} \times \text{PE} \times \text{MPL} \times \text{NP}$$

where FE = frequency of exposure
 PE = probability of the event
 MPL = maximum probable loss
 NP = number of persons at risk

MPL is a straightforward factor to assess — if a pole was to fall on a person, the outcome is assumed to be serious injury or death. For the purpose of the calculation therefore, this can be considered close to or equal to 1.0. The NP is a combination of the public access and the BT access — both these affect the number of people at risk. The other factors in the risk assessment were split between FE and PE as appropriate. The risk assessment methodology is consistent with the industry standard. It enables shallow poles to be assessed against a range of criteria and have scores attributed. These scores are then used to calculate a hazard rating number for the pole. If the HRN for the pole is above an agreed threshold, it is marked as defective (D) in the normal way. If it is below the agreed threshold, then it is given the new marking of shallow defective (SD). A shallow defective pole is a pole which, although it does not meet the documented depth standards, is not going to fall over and may be left in the ground until the next pole test cycle or until other circumstances warrant its replacement.

Two difficulties emerged which required resolution. Firstly, the HHT was unable to cope with the number range created by this multiplication technique, and, secondly, the pass/fail threshold had to be determined.

To deal with the first problem, the weighting factors were combined logarithmically. Thus, $\log_{10} \text{HRN}$ gives results in the range 0 to 14 rather than 1 to 10^{14} while maintaining the correct relationship.

The second problem — to determine the threshold — was solved by empirical research in the form of a threshold survey. A survey of 753 shallow poles was carried out by experienced pole testers over a three-week period. In this survey, the pole testers conducted their own experience-based risk assessments, as before, and made a judgement on whether the pole was D or SD, recording the result. They then carried out the formal BT risk assessment, the outcome of which was masked so as not to influence the views of the pole tester. The key results are represented in the pie chart in Fig 3. Of the 753 poles tested some turned out not to be shallow and others had no markings so could not be assessed, which left 689. The chart shows the percentages of those poles broken down into the following categories:

- agree 'SD' — these are poles that the model and tester both rate 'SD',

- agree 'D' — these are poles that the model and tester both rate 'D',
- disagree tester 'D' — these poles the testers rated 'D' but the model rated 'SD',
- disagree tester 'SD' — these poles the testers rated 'SD' but the model rated 'D',
- other reasons for 'D' — poles where the testers reported other factors, such as decay, etc.

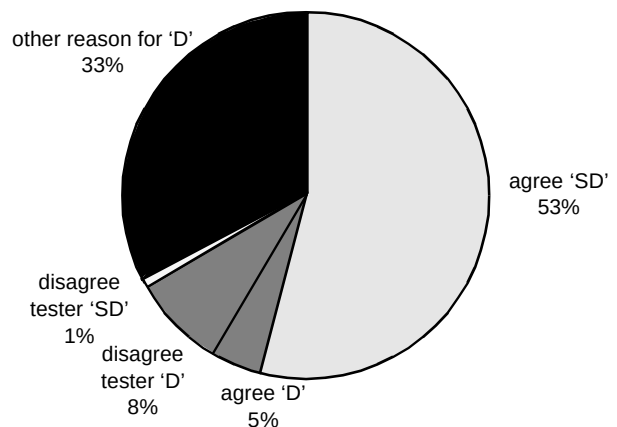


Fig 3 Correlation between model result and pole tester opinion.

The threshold survey showed that the risk assessment model provided a good, but not perfect, fit to the testers' opinions, 91% of poles tested were either 'agree SD', 'D for other reasons' or 'agree D', thus leaving a 9% area of concern. In 1% of cases the model had marked the pole 'D' while the pole tester had marked it 'SD'. This is acceptable as it shows the model applying a more rigid standard than the pole tester, i.e. fail safe. Detailed analysis of the remaining 8% identified a critical difference between the model and the pole tester's opinion relating to the ground profile near the pole, e.g. a ditch, bank. Although this was not severe enough to fail the pole in itself, it did affect the pole tester's view of the pole. The risk assessment model was modified on the basis of this new understanding, resulting in a much better fit between the model and the pole tester's view. This still left a number of poles which were marked D by the pole tester and SD by the model. In each instance the pole tester was interviewed and where appropriate a further joint site visit to the pole carried out. This work identified the reason for discrepancy as being primarily commercial judgement, e.g. new pole fixtures required. The pole tester had taken the view that it would probably be as easy and cost-effective to replace the particular pole. Having thus established that the model had in fact captured the experience-based risk assessment of the

pole testers, the project team felt confident enough to move to a larger scale technical trial. This trial would not only further validate the risk assessment model, but also provide an opportunity to test out the new data collection hardware and software.

4. Hand-held terminal

Research was carried out to identify the best solution for replacement of the existing hand-held terminal and its associated database. The solution chosen would have to address the shortcomings of the existing terminal, together with supporting data upload over the telephone network by modem. Following evaluation the Psion Workabout was identified as offering the best mix of usability, reliability and cost.

It was decided to replace the existing database with a new purpose-built database, designed following an extensive customer consultation exercise. This combination of a new, more powerful terminal and purpose-designed database provides greatly enhanced management information to enable better control of the programme.

5. Implementation

The output of the development project went to trial in two selected parts of the country — Scotland and the northern Home Counties. A total of 1264 poles were tested over the three-week trial period and the results calculated. As shallow poles had been a much more marked problem in Scotland than in northern Home Counties, the Scotland result is shown alone here for clarity (Fig 4).

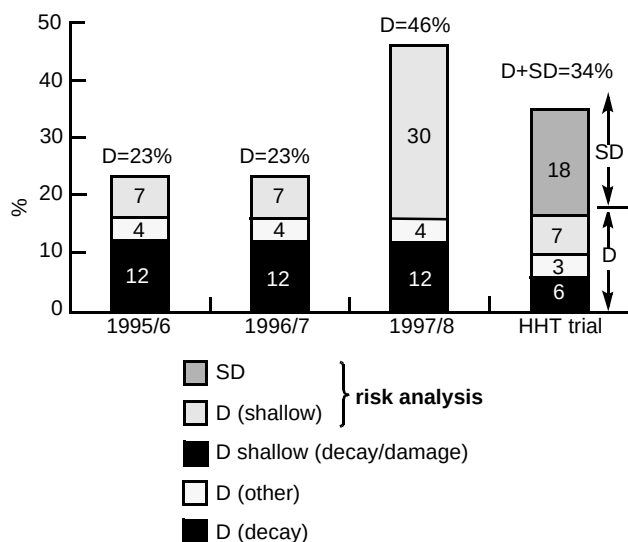


Fig 4 The reduction of the 'D'-pole find rate after applying the 'SD' category.

In this example, the find rate for defective poles over the last three years has sharply increased to 46%. The result from the trial reduces this to 16%.

The trial produced total concurrence between the risk assessment output and the pole tester's opinion. This concurrence is validation of the research and development work that went into capturing and documenting the informal risk analysis thought process of experienced pole testers and technical experts. It also validates the agreed threshold as being near to optimum. The present threshold setting has therefore been maintained for the pilot scheme and subsequent roll-out.

These trial results suggest that a significant percentage of poles found shallow do not require immediate replacement, thus enabling more defective poles to be scheduled for replacement sooner. The sample had a degree of regional diversity and was statistically valid although a much larger sample size is required before substantive claims can be made.

The hand-held terminal performed well in the operational environment. Data upload via modem was carried out on a daily basis without incident. A number of minor amendments have been captured as a consequence of operator feedback and these have now been incorporated in the terminal.

6. Conclusions

The progress of this work has validated the approach of applying a risk-assessment methodology to the prioritisation of work programmes in the access network and that by so doing the exposure of BT people and members of the public to risk can be reduced. Additionally the concept of using hand-held terminals for this purpose in the pole-testing environment is fully validated.

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Martin Redstall joined BT Laboratories in 1973, analysing epitaxial III-V semiconductors, after graduating with a BA in Physics (and later an MSc) from Essex University. Subsequently he worked on the reliability of semiconductor plastic packaging, IR semiconductor lasers and passive optical splitters.

He was a member of a European RACE project concerned with the reliability of optical splitters and the overall reliability of fibre networks, and subsequently worked on advanced methods and tools for planning fibre access networks. He presently leads a project using risk assessment techniques to optimise external plant expenditure while maintaining or improving safety in the access network.



Matthew Thomas graduated from Bradford University in 1985 with a degree in Mechanical Engineering, and in 1986 with an MSc in Microcomputer Systems. After finishing his industrial training with Wimpey Engineering, he carried out industrially sponsored research in the field of plant condition monitoring, gaining his doctorate from Manchester University in 1990. Subsequent to this he worked for an engineering consultancy in the field of maintenance management primarily in the petrochemical industry (on and offshore) prior to becoming a lecturer at Strathclyde

University where he specialised in condition monitoring and maintenance management. He joined BT Laboratories in 1995 to work on plant failure analysis and modelling to provide tools and information which can be used by the business to assist operational and strategic decision making, and is currently responsible for a number of teams working in this and other areas.



Wesley McKee is the national development manager for the asset assurance programme. He joined BT in 1973 as an apprentice and worked in a variety of roles including external planning before moving to BT's training college at Stone in the mid 1980s.

Since leaving training he has held a number of roles in technical support, process development and contracting, having introduced the concept of contractor technician accreditation to the access environment.

A masters graduate from Manchester Metropolitan University, his current academic interests include the development and motivation of geographically dispersed work groups.



Mike Halls joined BT in 1979 as a technician. Early work covered all aspects of fibre optic research. After graduating in 1988 with a BSc (Hons) in Electronics and Telecommunications from Essex University he joined a team investigating the application of ATM to passive optical networks. During this time he was responsible for both high-speed logic and optical interface design. This was followed by a period of involvement in technical trials and pilots. This included the LLOFT field trial at Bishops Cleeve, support to the TPON trial and pilot and more recently the collaborative BT/ Nortel ADSL trial. He is

currently working on a number of initiatives to assist the asset assurance programme in improving both the safety and cost effectiveness of the overhead network.